Newman-Penrose Formalism Calculations

Stephani et al., Equation (17.5_e1)

Metric

$ds^2 = -n\mathrm{d}x \otimes \mathrm{d}x + m\mathrm{d}x \otimes \mathrm{d}y + m\mathrm{d}y \otimes \mathrm{d}x + l\mathrm{d}y \otimes \mathrm{d}y + e^M \mathrm{d}z \otimes \mathrm{d}z + e^M \mathrm{d}t \otimes \mathrm{d}t$

Coordinates

$(x^\mu) = (x, y, z, t)$

Metric Functions

$n(t, z) = n$

$m(t, z) = m$

$l(t, z) = l$

$M(t, z) = M$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\sqrt{2}\sqrt{n}\right) dx + \left(\frac{1}{2}\sqrt{2}\sqrt{\frac{m^2}{n} + l} - \frac{\sqrt{2}m}{2\sqrt{n}}\right) dy$$

$$n = \left(\frac{1}{2}\sqrt{2}\sqrt{n}\right) dx + \left(-\frac{1}{2}\sqrt{2}\sqrt{\frac{m^2}{n} + l} - \frac{\sqrt{2}m}{2\sqrt{n}}\right) dy$$

$$m = \left(\frac{1}{2}\,\sqrt{2}e^{\left(\frac{1}{2}\,M\right)}\right)\,dz + \left(\frac{1}{2}i\,\sqrt{2}e^{\left(\frac{1}{2}\,M\right)}\right)\,dt$$

$$\overline{m} = \left(\frac{1}{2}\,\sqrt{2}e^{\left(\frac{1}{2}\,M\right)}\right)\,dz + \left(-\frac{1}{2}i\,\sqrt{2}e^{\left(\frac{1}{2}\,M\right)}\right)\,dt$$

Contravariant components

$$l = \left(-\frac{m^2}{\left(\sqrt{2}m^2 + \sqrt{2}ln\right)\sqrt{n}} - \frac{l\sqrt{n}}{\sqrt{2}m^2 + \sqrt{2}ln} + \frac{\sqrt{m^2 + ln}m}{\left(\sqrt{2}m^2 + \sqrt{2}ln\right)\sqrt{n}}\right)\,dx + \left(\frac{\sqrt{m^2 + ln}\sqrt{n}}{\sqrt{2}m^2 + \sqrt{2}ln}\right)\,dy$$

$$n = \left(-\frac{m^2}{\left(\sqrt{2}m^2 + \sqrt{2}ln\right)\sqrt{n}} - \frac{l\sqrt{n}}{\sqrt{2}m^2 + \sqrt{2}ln} - \frac{\sqrt{m^2 + ln}m}{\left(\sqrt{2}m^2 + \sqrt{2}ln\right)\sqrt{n}}\right)\,dx + \left(-\frac{\sqrt{m^2 + ln}\sqrt{n}}{\sqrt{2}m^2 + \sqrt{2}ln}\right)\,dy$$

$$m = \left(\frac{1}{2}\,\sqrt{2}e^{\left(-\frac{1}{2}\,M\right)}\right)\,dz + \left(\frac{1}{2}i\,\sqrt{2}e^{\left(-\frac{1}{2}\,M\right)}\right)\,dt$$

$$\overline{m} = \left(\frac{1}{2}\,\sqrt{2}e^{\left(-\frac{1}{2}\,M\right)}\right)\,dz + \left(-\frac{1}{2}i\,\sqrt{2}e^{\left(-\frac{1}{2}\,M\right)}\right)\,dt$$

Directional Derivatives

$$D = -\frac{m^2}{\left(\sqrt{2}m^2 + \sqrt{2}ln\right)\sqrt{n}} - \frac{l\sqrt{n}}{\sqrt{2}m^2 + \sqrt{2}ln} + \frac{\sqrt{m^2 + ln}m}{\left(\sqrt{2}m^2 + \sqrt{2}ln\right)\sqrt{n}}\,\partial_x + \frac{\sqrt{m^2 + ln}\sqrt{n}}{\sqrt{2}m^2 + \sqrt{2}ln}\,\partial_y$$

$$\Delta = -\frac{m^2}{\left(\sqrt{2}m^2 + \sqrt{2}ln\right)\sqrt{n}} - \frac{l\sqrt{n}}{\sqrt{2}m^2 + \sqrt{2}ln} - \frac{\sqrt{m^2 + ln}m}{\left(\sqrt{2}m^2 + \sqrt{2}ln\right)\sqrt{n}}\,\partial_x - \frac{\sqrt{m^2 + ln}\sqrt{n}}{\sqrt{2}m^2 + \sqrt{2}ln}\,\partial_y$$

$$\delta = \frac{1}{2}\,\sqrt{2}e^{\left(-\frac{1}{2}\,M\right)}\,\partial_z + \frac{1}{2}i\,\sqrt{2}e^{\left(-\frac{1}{2}\,M\right)}\,\partial_t$$

$$\overline{\delta} = \frac{1}{2}\sqrt{2}e^{\left(-\frac{1}{2}M\right)}\partial_z - \frac{1}{2}i\sqrt{2}e^{\left(-\frac{1}{2}M\right)}\partial_t$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\begin{aligned}\kappa = {} & \frac{i\sqrt{2}n^2\frac{\partial}{\partial t}l}{8\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} + \frac{\sqrt{2}n^2\frac{\partial}{\partial z}l}{8\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} + \frac{i\sqrt{2}mn\frac{\partial}{\partial t}m}{4\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} + \frac{\sqrt{2}mn\frac{\partial}{\partial z}m}{4\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} \\ & - \frac{i\sqrt{2}m^2\frac{\partial}{\partial t}n}{4\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} - \frac{i\sqrt{2}ln\frac{\partial}{\partial t}n}{8\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} - \frac{\sqrt{2}m^2\frac{\partial}{\partial z}n}{4\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} - \frac{\sqrt{2}ln\frac{\partial}{\partial z}n}{8\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} \\ & - \frac{i\sqrt{2}\sqrt{m^2+ln}n\frac{\partial}{\partial t}m}{4\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} - \frac{\sqrt{2}\sqrt{m^2+ln}n\frac{\partial}{\partial z}m}{4\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} + \frac{i\sqrt{2}\sqrt{m^2+ln}m\frac{\partial}{\partial t}n}{4\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} + \frac{\sqrt{2}\sqrt{m^2+ln}m\frac{\partial}{\partial z}n}{4\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)}\end{aligned}$$

$$\mu = 0$$

$$\lambda = 0$$

$$\begin{aligned}\nu = {} & \frac{i\sqrt{2}n^2\frac{\partial}{\partial t}l}{8\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} - \frac{\sqrt{2}n^2\frac{\partial}{\partial z}l}{8\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} + \frac{i\sqrt{2}mn\frac{\partial}{\partial t}m}{4\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} - \frac{\sqrt{2}mn\frac{\partial}{\partial z}m}{4\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} \\ & - \frac{i\sqrt{2}m^2\frac{\partial}{\partial t}n}{4\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} - \frac{i\sqrt{2}ln\frac{\partial}{\partial t}n}{8\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} + \frac{\sqrt{2}m^2\frac{\partial}{\partial z}n}{4\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} + \frac{\sqrt{2}ln\frac{\partial}{\partial z}n}{8\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} \\ & + \frac{i\sqrt{2}\sqrt{m^2+ln}n\frac{\partial}{\partial t}m}{4\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} - \frac{\sqrt{2}\sqrt{m^2+ln}n\frac{\partial}{\partial z}m}{4\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} - \frac{i\sqrt{2}\sqrt{m^2+ln}m\frac{\partial}{\partial t}n}{4\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} + \frac{\sqrt{2}\sqrt{m^2+ln}m\frac{\partial}{\partial z}n}{4\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)}\end{aligned}$$

$$\begin{aligned}\alpha = {} & \frac{i\sqrt{2}m^2n\frac{\partial}{\partial t}M}{8\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} + \frac{i\sqrt{2}ln^2\frac{\partial}{\partial t}M}{8\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} - \frac{\sqrt{2}m^2n\frac{\partial}{\partial z}M}{8\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} - \frac{\sqrt{2}ln^2\frac{\partial}{\partial z}M}{8\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} \\ & + \frac{i\sqrt{2}\sqrt{m^2+ln}n\frac{\partial}{\partial t}m}{8\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} - \frac{\sqrt{2}\sqrt{m^2+ln}n\frac{\partial}{\partial z}m}{8\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} - \frac{i\sqrt{2}\sqrt{m^2+ln}m\frac{\partial}{\partial t}n}{8\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} + \frac{\sqrt{2}\sqrt{m^2+ln}m\frac{\partial}{\partial z}n}{8\left(e^{\left(\frac{1}{2}M\right)}m^2n + e^{\left(\frac{1}{2}M\right)}ln^2\right)}\end{aligned}$$

$$\beta = \frac{i\sqrt{2}m^2 n \frac{\partial}{\partial t}M}{8\left(e^{\left(\frac{1}{2}M\right)}m^2 n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} + \frac{i\sqrt{2}ln^2 \frac{\partial}{\partial t}M}{8\left(e^{\left(\frac{1}{2}M\right)}m^2 n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} + \frac{\sqrt{2}m^2 n \frac{\partial}{\partial z}M}{8\left(e^{\left(\frac{1}{2}M\right)}m^2 n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} + \frac{\sqrt{2}ln^2 \frac{\partial}{\partial z}M}{8\left(e^{\left(\frac{1}{2}M\right)}m^2 n + e^{\left(\frac{1}{2}M\right)}ln^2\right)}$$
$$- \frac{i\sqrt{2}\sqrt{m^2+ln}n \frac{\partial}{\partial t}m}{8\left(e^{\left(\frac{1}{2}M\right)}m^2 n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} - \frac{\sqrt{2}\sqrt{m^2+ln}n \frac{\partial}{\partial z}m}{8\left(e^{\left(\frac{1}{2}M\right)}m^2 n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} + \frac{i\sqrt{2}\sqrt{m^2+ln}m \frac{\partial}{\partial t}n}{8\left(e^{\left(\frac{1}{2}M\right)}m^2 n + e^{\left(\frac{1}{2}M\right)}ln^2\right)} + \frac{\sqrt{2}\sqrt{m^2+ln}m \frac{\partial}{\partial z}n}{8\left(e^{\left(\frac{1}{2}M\right)}m^2 n + e^{\left(\frac{1}{2}M\right)}ln^2\right)}$$

$$\tau = -\frac{i\sqrt{2}n \frac{\partial}{\partial t}l}{8\left(e^{\left(\frac{1}{2}M\right)}m^2 + e^{\left(\frac{1}{2}M\right)}ln\right)} - \frac{\sqrt{2}n \frac{\partial}{\partial z}l}{8\left(e^{\left(\frac{1}{2}M\right)}m^2 + e^{\left(\frac{1}{2}M\right)}ln\right)} - \frac{i\sqrt{2}m \frac{\partial}{\partial t}m}{4\left(e^{\left(\frac{1}{2}M\right)}m^2 + e^{\left(\frac{1}{2}M\right)}ln\right)} - \frac{\sqrt{2}m \frac{\partial}{\partial z}m}{4\left(e^{\left(\frac{1}{2}M\right)}m^2 + e^{\left(\frac{1}{2}M\right)}ln\right)} - \frac{i\sqrt{2}l \frac{\partial}{\partial t}n}{8\left(e^{\left(\frac{1}{2}M\right)}m^2 + e^{\left(\frac{1}{2}M\right)}ln\right)} - \frac{\sqrt{2}l \frac{\partial}{\partial z}n}{8\left(e^{\left(\frac{1}{2}M\right)}m^2 + e^{\left(\frac{1}{2}M\right)}ln\right)}$$

$$\pi = -\frac{i\sqrt{2}n \frac{\partial}{\partial t}l}{8\left(e^{\left(\frac{1}{2}M\right)}m^2 + e^{\left(\frac{1}{2}M\right)}ln\right)} + \frac{\sqrt{2}n \frac{\partial}{\partial z}l}{8\left(e^{\left(\frac{1}{2}M\right)}m^2 + e^{\left(\frac{1}{2}M\right)}ln\right)} - \frac{i\sqrt{2}m \frac{\partial}{\partial t}m}{4\left(e^{\left(\frac{1}{2}M\right)}m^2 + e^{\left(\frac{1}{2}M\right)}ln\right)} + \frac{\sqrt{2}m \frac{\partial}{\partial z}m}{4\left(e^{\left(\frac{1}{2}M\right)}m^2 + e^{\left(\frac{1}{2}M\right)}ln\right)} - \frac{i\sqrt{2}l \frac{\partial}{\partial t}n}{8\left(e^{\left(\frac{1}{2}M\right)}m^2 + e^{\left(\frac{1}{2}M\right)}ln\right)} + \frac{\sqrt{2}l \frac{\partial}{\partial z}n}{8\left(e^{\left(\frac{1}{2}M\right)}m^2 + e^{\left(\frac{1}{2}M\right)}ln\right)}$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$
\begin{aligned}
\Phi_{00} = {} & \frac{n^3 \frac{\partial}{\partial t} l^2}{16\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{m^2 n^2 \frac{\partial^2}{(\partial t)^2} l}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{l n^3 \frac{\partial^2}{(\partial t)^2} l}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{n^3 \frac{\partial}{\partial z} l^2}{16\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} \\
& - \frac{m^2 n^2 \frac{\partial^2}{(\partial z)^2} l}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{l n^3 \frac{\partial^2}{(\partial z)^2} l}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{m n^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial t} m}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{m^2 n \frac{\partial}{\partial t} m^2}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} \\
& - \frac{m^3 n \frac{\partial^2}{(\partial t)^2} m}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{l m n^2 \frac{\partial^2}{(\partial t)^2} m}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{m n^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{m^2 n \frac{\partial}{\partial z} m^2}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} \\
& - \frac{m^3 n \frac{\partial^2}{(\partial z)^2} m}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{l m n^2 \frac{\partial^2}{(\partial z)^2} m}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{m^2 n \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{m^3 \frac{\partial}{\partial t} m \frac{\partial}{\partial t} n}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} \\
& - \frac{l m^2 \frac{\partial}{\partial t} n^2}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{l^2 n \frac{\partial}{\partial t} n^2}{16\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{m^4 \frac{\partial^2}{(\partial t)^2} n}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} \\
& + \frac{3\,l m^2 n \frac{\partial^2}{(\partial t)^2} n}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{l^2 n^2 \frac{\partial^2}{(\partial t)^2} n}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{m^2 n \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{m^3 \frac{\partial}{\partial z} m \frac{\partial}{\partial z} n}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} \\
& - \frac{l m^2 \frac{\partial}{\partial z} n^2}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{l^2 n \frac{\partial}{\partial z} n^2}{16\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{m^4 \frac{\partial^2}{(\partial z)^2} n}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{3\,l m^2 n \frac{\partial^2}{(\partial z)^2} n}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} \\
& + \frac{l^2 n^2 \frac{\partial^2}{(\partial z)^2} n}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{\sqrt{m^2 + l n}\, n^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial t} m}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{\sqrt{m^2 + l n}\, m n \frac{\partial}{\partial t} m^2}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{\sqrt{m^2 + l n}\, m^2 n \frac{\partial^2}{(\partial t)^2} m}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} \\
& + \frac{\sqrt{m^2 + l n}\, l n^2 \frac{\partial^2}{(\partial t)^2} m}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{\sqrt{m^2 + l n}\, n^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{\sqrt{m^2 + l n}\, m n \frac{\partial}{\partial z} m^2}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \dots
\end{aligned}
$$

$$
\Phi_{01} = 0
$$

$$
\begin{aligned}
\Phi_{02} = &-\frac{m^2 n \frac{\partial}{\partial t} M \frac{\partial}{\partial t} l}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} - \frac{l n^2 \frac{\partial}{\partial t} M \frac{\partial}{\partial t} l}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} + \frac{i\, m^2 n \frac{\partial}{\partial z} M \frac{\partial}{\partial t} l}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} + \frac{i\, l n^2 \frac{\partial}{\partial z} M \frac{\partial}{\partial t} l}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} \\
&+ \frac{i\, m^2 n \frac{\partial}{\partial t} M \frac{\partial}{\partial z} l}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} + \frac{i\, l n^2 \frac{\partial}{\partial t} M \frac{\partial}{\partial z} l}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} + \frac{m^2 n \frac{\partial}{\partial z} M \frac{\partial}{\partial z} l}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} \\
&+ \frac{l n^2 \frac{\partial}{\partial z} M \frac{\partial}{\partial z} l}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} - \frac{m^3 \frac{\partial}{\partial t} M \frac{\partial}{\partial t} m}{4\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} - \frac{l m n \frac{\partial}{\partial t} M \frac{\partial}{\partial t} m}{4\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} + \frac{i\, m^3 \frac{\partial}{\partial z} M \frac{\partial}{\partial t} m}{4\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} \\
&+ \frac{i\, l m n \frac{\partial}{\partial z} M \frac{\partial}{\partial t} m}{4\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} + \frac{i\, m^3 \frac{\partial}{\partial t} M \frac{\partial}{\partial z} m}{4\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} + \frac{i\, l m n \frac{\partial}{\partial t} M \frac{\partial}{\partial z} m}{4\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} + \frac{m^3 \frac{\partial}{\partial z} M \frac{\partial}{\partial z} m}{4\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} \\
&+ \frac{l m n \frac{\partial}{\partial z} M \frac{\partial}{\partial z} m}{4\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} - \frac{l m^2 \frac{\partial}{\partial t} M \frac{\partial}{\partial t} n}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} - \frac{l^2 n \frac{\partial}{\partial t} M \frac{\partial}{\partial t} n}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} + \frac{i\, l m^2 \frac{\partial}{\partial z} M \frac{\partial}{\partial t} n}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} \\
&+ \frac{i\, l^2 n \frac{\partial}{\partial z} M \frac{\partial}{\partial t} n}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} + \frac{i\, l m^2 \frac{\partial}{\partial t} M \frac{\partial}{\partial z} n}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} + \frac{i\, l^2 n \frac{\partial}{\partial t} M \frac{\partial}{\partial z} n}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} + \frac{l m^2 \frac{\partial}{\partial z} M \frac{\partial}{\partial z} n}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} \\
&+ \frac{l^2 n \frac{\partial}{\partial z} M \frac{\partial}{\partial z} n}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} - \frac{n^2 \frac{\partial}{\partial t} l^2}{16\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} + \frac{m^2 n \frac{\partial^2}{(\partial t)^2} l}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} \\
&+ \frac{l n^2 \frac{\partial^2}{(\partial t)^2} l}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} - \frac{i\, m^2 n \frac{\partial^2}{\partial t \partial z} l}{4\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} - \frac{i\, l n^2 \frac{\partial^2}{\partial t \partial z} l}{4\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} + \frac{i\, n^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} l}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} \\
&+ \frac{n^2 \frac{\partial}{\partial z} l^2}{16\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} - \frac{m^2 n \frac{\partial^2}{(\partial z)^2} l}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} - \frac{l n^2 \frac{\partial^2}{(\partial z)^2} l}{8\left(e^M m^4 + 2\, e^M l m^2 n + e^M l^2 n^2\right)} - \ldots
\end{aligned}
$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{m^2 \frac{\partial^2}{(\partial t)^2} M}{8\left(e^M m^2 + e^M l n\right)} - \frac{l n \frac{\partial^2}{(\partial t)^2} M}{8\left(e^M m^2 + e^M l n\right)} - \frac{m^2 \frac{\partial^2}{(\partial z)^2} M}{8\left(e^M m^2 + e^M l n\right)} - \frac{l n \frac{\partial^2}{(\partial z)^2} M}{8\left(e^M m^2 + e^M l n\right)} + \frac{\frac{\partial}{\partial t} m^2}{16\left(e^M m^2 + e^M l n\right)} + \frac{\frac{\partial}{\partial z} m^2}{16\left(e^M m^2 + e^M l n\right)} + \frac{\frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{16\left(e^M m^2 + e^M l n\right)} + \frac{\frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{16\left(e^M m^2 + e^M l n\right)}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = -\frac{m^2 n \frac{\partial}{\partial t} M \frac{\partial}{\partial t} l}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} - \frac{l n^2 \frac{\partial}{\partial t} M \frac{\partial}{\partial t} l}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} - \frac{i\, m^2 n \frac{\partial}{\partial z} M \frac{\partial}{\partial t} l}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} - \frac{i\, l n^2 \frac{\partial}{\partial z} M \frac{\partial}{\partial t} l}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)}$$
$$- \frac{i\, m^2 n \frac{\partial}{\partial t} M \frac{\partial}{\partial z} l}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} - \frac{i\, l n^2 \frac{\partial}{\partial t} M \frac{\partial}{\partial z} l}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} + \frac{m^2 n \frac{\partial}{\partial z} M \frac{\partial}{\partial z} l}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} + \frac{l n^2 \frac{\partial}{\partial z} M \frac{\partial}{\partial z} l}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)}$$
$$- \frac{m^3 \frac{\partial}{\partial t} M \frac{\partial}{\partial t} m}{4\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} - \frac{l m n \frac{\partial}{\partial t} M \frac{\partial}{\partial t} m}{4\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} - \frac{i\, m^3 \frac{\partial}{\partial z} M \frac{\partial}{\partial t} m}{4\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} - \frac{i\, l m n \frac{\partial}{\partial z} M \frac{\partial}{\partial t} m}{4\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)}$$
$$- \frac{i\, m^3 \frac{\partial}{\partial t} M \frac{\partial}{\partial z} m}{4\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} - \frac{i\, l m n \frac{\partial}{\partial t} M \frac{\partial}{\partial z} m}{4\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} + \frac{m^3 \frac{\partial}{\partial z} M \frac{\partial}{\partial z} m}{4\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} + \frac{l m n \frac{\partial}{\partial z} M \frac{\partial}{\partial z} m}{4\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)}$$
$$- \frac{l m^2 \frac{\partial}{\partial t} M \frac{\partial}{\partial t} n}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} - \frac{l^2 n \frac{\partial}{\partial t} M \frac{\partial}{\partial t} n}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} - \frac{i\, l m^2 \frac{\partial}{\partial z} M \frac{\partial}{\partial t} n}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} - \frac{i\, l^2 n \frac{\partial}{\partial z} M \frac{\partial}{\partial t} n}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)}$$
$$- \frac{i\, l m^2 \frac{\partial}{\partial t} M \frac{\partial}{\partial z} n}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} - \frac{i\, l^2 n \frac{\partial}{\partial t} M \frac{\partial}{\partial z} n}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} + \frac{l m^2 \frac{\partial}{\partial z} M \frac{\partial}{\partial z} n}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} + \frac{l^2 n \frac{\partial}{\partial z} M \frac{\partial}{\partial z} n}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)}$$
$$- \frac{n^2 \frac{\partial}{\partial t} l^2}{16\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} + \frac{m^2 n \frac{\partial^2}{(\partial t)^2} l}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} + \frac{l n^2 \frac{\partial^2}{(\partial t)^2} l}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)}$$
$$+ \frac{i\, m^2 n \frac{\partial^2}{\partial t \partial z} l}{4\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} + \frac{i\, l n^2 \frac{\partial^2}{\partial t \partial z} l}{4\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} - \frac{i\, n^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} l}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)}$$
$$+ \frac{n^2 \frac{\partial}{\partial z} l^2}{16\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} - \frac{m^2 n \frac{\partial^2}{(\partial z)^2} l}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} - \frac{l n^2 \frac{\partial^2}{(\partial z)^2} l}{8\left(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2\right)} - \dots$$

$$\Phi_{21} = 0$$

$$
\begin{aligned}
\Phi_{22} = {} & \frac{n^3 \frac{\partial}{\partial t} l^2}{16\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{m^2 n^2 \frac{\partial^2}{(\partial t)^2} l}{8\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{l n^3 \frac{\partial^2}{(\partial t)^2} l}{8\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{n^3 \frac{\partial}{\partial z} l^2}{16\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} \\
& - \frac{m^2 n^2 \frac{\partial^2}{(\partial z)^2} l}{8\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{l n^3 \frac{\partial^2}{(\partial z)^2} l}{8\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{m n^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial t} m}{4\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{m^2 n \frac{\partial}{\partial t} m^2}{4\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} \\
& - \frac{m^3 n \frac{\partial^2}{(\partial t)^2} m}{4\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{l m n^2 \frac{\partial^2}{(\partial t)^2} m}{4\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{m n^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{4\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{m^2 n \frac{\partial}{\partial z} m^2}{4\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} \\
& - \frac{m^3 n \frac{\partial^2}{(\partial z)^2} m}{4\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{l m n^2 \frac{\partial^2}{(\partial z)^2} m}{4\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{m^2 n \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{8\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{m^3 \frac{\partial}{\partial t} m \frac{\partial}{\partial t} n}{4\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} \\
& - \frac{l m^2 \frac{\partial}{\partial t} n^2}{8\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{l^2 n \frac{\partial}{\partial t} n^2}{16\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{m^4 \frac{\partial^2}{(\partial t)^2} n}{4\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} \\
& + \frac{3\, l m^2 n \frac{\partial^2}{(\partial t)^2} n}{8\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{l^2 n^2 \frac{\partial^2}{(\partial t)^2} n}{8\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{m^2 n \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{8\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{m^3 \frac{\partial}{\partial z} m \frac{\partial}{\partial z} n}{4\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} \\
& - \frac{l m^2 \frac{\partial}{\partial z} n^2}{8\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{l^2 n \frac{\partial}{\partial z} n^2}{16\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{m^4 \frac{\partial^2}{(\partial z)^2} n}{4\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{3\, l m^2 n \frac{\partial^2}{(\partial z)^2} n}{8\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} \\
& + \frac{l^2 n^2 \frac{\partial^2}{(\partial z)^2} n}{8\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{\sqrt{m^2 + l n}\, n^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial t} m}{8\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{\sqrt{m^2 + l n}\, m n \frac{\partial}{\partial t} m^2}{4\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{\sqrt{m^2 + l n}\, m^2 n \frac{\partial^2}{(\partial t)^2} m}{4\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} \\
& - \frac{\sqrt{m^2 + l n}\, l n^2 \frac{\partial^2}{(\partial t)^2} m}{4\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{\sqrt{m^2 + l n}\, n^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{8\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{\sqrt{m^2 + l n}\, m n \frac{\partial}{\partial z} m^2}{4\left(e^M m^4 n + 2\, e^M l m^2 n^2 + e^M l^2 n^3\right)} - \ldots
\end{aligned}
$$

$$
\begin{aligned}
\Lambda = &-\frac{m^4 \frac{\partial^2}{(\partial t)^2} M}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{l m^2 n \frac{\partial^2}{(\partial t)^2} M}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{l^2 n^2 \frac{\partial^2}{(\partial t)^2} M}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{m^4 \frac{\partial^2}{(\partial z)^2} M}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} \\
&- \frac{l m^2 n \frac{\partial^2}{(\partial z)^2} M}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{l^2 n^2 \frac{\partial^2}{(\partial z)^2} M}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{n^2 \frac{\partial}{\partial t} l^2}{48\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{m^2 n \frac{\partial^2}{(\partial t)^2} l}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} \\
&- \frac{l n^2 \frac{\partial^2}{(\partial t)^2} l}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{n^2 \frac{\partial}{\partial z} l^2}{48\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{m^2 n \frac{\partial^2}{(\partial z)^2} l}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{l n^2 \frac{\partial^2}{(\partial z)^2} l}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} \\
&+ \frac{m n \frac{\partial}{\partial t} l \frac{\partial}{\partial t} m}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{m^2 \frac{\partial}{\partial t} m^2}{48\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{l n \frac{\partial}{\partial t} m^2}{16\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} \\
&- \frac{m^3 \frac{\partial^2}{(\partial t)^2} m}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{l m n \frac{\partial^2}{(\partial t)^2} m}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{m n \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{m^2 \frac{\partial}{\partial z} m^2}{48\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} \\
&- \frac{l n \frac{\partial}{\partial z} m^2}{16\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{m^3 \frac{\partial^2}{(\partial z)^2} m}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{l m n \frac{\partial^2}{(\partial z)^2} m}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{m^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{16\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} \\
&- \frac{l n \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{48\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{l m \frac{\partial}{\partial t} m \frac{\partial}{\partial t} n}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{l^2 \frac{\partial}{\partial t} n^2}{48\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} \\
&- \frac{l m^2 \frac{\partial^2}{(\partial t)^2} n}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{l^2 n \frac{\partial^2}{(\partial t)^2} n}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{m^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{16\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{l n \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{48\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} \\
&+ \frac{l m \frac{\partial}{\partial z} m \frac{\partial}{\partial z} n}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{l^2 \frac{\partial}{\partial z} n^2}{48\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{l m^2 \frac{\partial^2}{(\partial z)^2} n}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{l^2 n \frac{\partial^2}{(\partial z)^2} n}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)}
\end{aligned}
$$

Weyl Tensor Components

$$
\begin{aligned}
\Psi_0 = &-\frac{m^2n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial t}l}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{l n^3\frac{\partial}{\partial t}M\frac{\partial}{\partial t}l}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{i\,m^2n^2\frac{\partial}{\partial z}M\frac{\partial}{\partial t}l}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{i\,l n^3\frac{\partial}{\partial z}M\frac{\partial}{\partial t}l}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)}\\
&+ \frac{i\,m^2n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial z}l}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{i\,l n^3\frac{\partial}{\partial t}M\frac{\partial}{\partial z}l}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{m^2n^2\frac{\partial}{\partial z}M\frac{\partial}{\partial z}l}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)}\\
&+ \frac{l n^3\frac{\partial}{\partial z}M\frac{\partial}{\partial z}l}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{m^3 n\frac{\partial}{\partial t}M\frac{\partial}{\partial t}m}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{l m n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial t}m}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{i\,m^3 n\frac{\partial}{\partial z}M\frac{\partial}{\partial t}m}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)}\\
&+ \frac{i\,l m n^2\frac{\partial}{\partial z}M\frac{\partial}{\partial t}m}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{i\,m^3 n\frac{\partial}{\partial t}M\frac{\partial}{\partial z}m}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{i\,l m n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial z}m}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{m^3 n\frac{\partial}{\partial z}M\frac{\partial}{\partial z}m}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)}\\
&+ \frac{l m n^2\frac{\partial}{\partial z}M\frac{\partial}{\partial z}m}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{m^4\frac{\partial}{\partial t}M\frac{\partial}{\partial t}n}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{3\,l m^2 n\frac{\partial}{\partial t}M\frac{\partial}{\partial t}n}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{l^2 n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial t}n}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)}\\
&- \frac{i\,m^4\frac{\partial}{\partial z}M\frac{\partial}{\partial t}n}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{3i\,l m^2 n\frac{\partial}{\partial z}M\frac{\partial}{\partial t}n}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{i\,l^2 n^2\frac{\partial}{\partial z}M\frac{\partial}{\partial t}n}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{i\,m^4\frac{\partial}{\partial t}M\frac{\partial}{\partial z}n}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)}\\
&- \frac{3i\,l m^2 n\frac{\partial}{\partial t}M\frac{\partial}{\partial z}n}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{i\,l^2 n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial z}n}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{m^4\frac{\partial}{\partial z}M\frac{\partial}{\partial z}n}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \frac{3\,l m^2 n\frac{\partial}{\partial z}M\frac{\partial}{\partial z}n}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)}\\
&- \frac{l^2 n^2\frac{\partial}{\partial z}M\frac{\partial}{\partial z}n}{8\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{\sqrt{m^2+ln}\,m^2 n\frac{\partial}{\partial t}M\frac{\partial}{\partial t}m}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} + \frac{\sqrt{m^2+ln}\,l n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial t}m}{4\left(e^M m^4 n + 2\,e^M l m^2 n^2 + e^M l^2 n^3\right)} - \cdots
\end{aligned}
$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{m^4 \frac{\partial^2}{(\partial t)^2} M}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{l m^2 n \frac{\partial^2}{(\partial t)^2} M}{6\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{l^2 n^2 \frac{\partial^2}{(\partial t)^2} M}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{m^4 \frac{\partial^2}{(\partial z)^2} M}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)}$$
$$+ \frac{l m^2 n \frac{\partial^2}{(\partial z)^2} M}{6\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{l^2 n^2 \frac{\partial^2}{(\partial z)^2} M}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{n^2 \frac{\partial}{\partial t} l^2}{48\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{m^2 n \frac{\partial^2}{(\partial t)^2} l}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)}$$
$$- \frac{l n^2 \frac{\partial^2}{(\partial t)^2} l}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{n^2 \frac{\partial}{\partial z} l^2}{48\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{m^2 n \frac{\partial^2}{(\partial z)^2} l}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{l n^2 \frac{\partial^2}{(\partial z)^2} l}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)}$$
$$+ \frac{m n \frac{\partial}{\partial t} l \frac{\partial}{\partial t} m}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{m^2 \frac{\partial}{\partial t} m^2}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{m^3 \frac{\partial^2}{(\partial t)^2} m}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{l m n \frac{\partial^2}{(\partial t)^2} m}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)}$$
$$+ \frac{m n \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{m^2 \frac{\partial}{\partial z} m^2}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{m^3 \frac{\partial^2}{(\partial z)^2} m}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{l m n \frac{\partial^2}{(\partial z)^2} m}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)}$$
$$+ \frac{l n \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{l m \frac{\partial}{\partial t} m \frac{\partial}{\partial t} n}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{l^2 \frac{\partial}{\partial t} n^2}{48\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{l m^2 \frac{\partial^2}{(\partial t)^2} n}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)}$$
$$- \frac{l^2 n \frac{\partial^2}{(\partial t)^2} n}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{l n \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{l m \frac{\partial}{\partial z} m \frac{\partial}{\partial z} n}{12\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{l^2 \frac{\partial}{\partial z} n^2}{48\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)}$$
$$- \frac{l m^2 \frac{\partial^2}{(\partial z)^2} n}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{l^2 n \frac{\partial^2}{(\partial z)^2} n}{24\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{i\,\sqrt{m^2 + l n} n \frac{\partial}{\partial z} l \frac{\partial}{\partial t} m}{8\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{i\,\sqrt{m^2 + l n} n \frac{\partial}{\partial t} l \frac{\partial}{\partial z} m}{8\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)}$$
$$- \frac{i\,\sqrt{m^2 + l n} m \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{8\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{i\,\sqrt{m^2 + l n} l \frac{\partial}{\partial z} m \frac{\partial}{\partial t} n}{8\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} + \frac{i\,\sqrt{m^2 + l n} m \frac{\partial}{\partial t} l \frac{\partial}{\partial z} n}{8\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)} - \frac{i\,\sqrt{m^2 + l n} l \frac{\partial}{\partial t} m \frac{\partial}{\partial z} n}{8\,(e^M m^4 + 2\,e^M l m^2 n + e^M l^2 n^2)}$$

$$\Psi_3 = 0$$

$$
\begin{aligned}
\Psi_4 = &-\frac{m^2n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial t}l}{8\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}-\frac{l n^3\frac{\partial}{\partial t}M\frac{\partial}{\partial t}l}{8\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}-\frac{i\,m^2n^2\frac{\partial}{\partial z}M\frac{\partial}{\partial t}l}{8\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}-\frac{i\,l n^3\frac{\partial}{\partial z}M\frac{\partial}{\partial t}l}{8\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}\\
&-\frac{i\,m^2n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial z}l}{8\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}-\frac{i\,l n^3\frac{\partial}{\partial t}M\frac{\partial}{\partial z}l}{8\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}+\frac{m^2n^2\frac{\partial}{\partial z}M\frac{\partial}{\partial z}l}{8\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}+\frac{l n^3\frac{\partial}{\partial z}M\frac{\partial}{\partial z}l}{8\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}\\
&-\frac{m^3n\frac{\partial}{\partial t}M\frac{\partial}{\partial t}m}{4\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}-\frac{l m n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial t}m}{4\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}-\frac{i\,m^3n\frac{\partial}{\partial z}M\frac{\partial}{\partial t}m}{4\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}-\frac{i\,l m n^2\frac{\partial}{\partial z}M\frac{\partial}{\partial t}m}{4\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}\\
&-\frac{i\,m^3n\frac{\partial}{\partial t}M\frac{\partial}{\partial z}m}{4\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}-\frac{i\,l m n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial z}m}{4\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}+\frac{m^3n\frac{\partial}{\partial z}M\frac{\partial}{\partial z}m}{4\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}+\frac{l m n^2\frac{\partial}{\partial z}M\frac{\partial}{\partial z}m}{4\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}\\
&+\frac{m^4\frac{\partial}{\partial t}M\frac{\partial}{\partial t}n}{4\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}+\frac{3\,l m^2 n\frac{\partial}{\partial t}M\frac{\partial}{\partial t}n}{8\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}+\frac{l^2n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial t}n}{8\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}\\
&+\frac{i\,m^4\frac{\partial}{\partial z}M\frac{\partial}{\partial t}n}{4\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}+\frac{3i\,l m^2 n\frac{\partial}{\partial z}M\frac{\partial}{\partial t}n}{8\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}+\frac{i\,l^2n^2\frac{\partial}{\partial z}M\frac{\partial}{\partial t}n}{8\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}+\frac{i\,m^4\frac{\partial}{\partial t}M\frac{\partial}{\partial z}n}{4\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}\\
&+\frac{3i\,l m^2 n\frac{\partial}{\partial t}M\frac{\partial}{\partial z}n}{8\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}+\frac{i\,l^2n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial z}n}{8\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}-\frac{m^4\frac{\partial}{\partial z}M\frac{\partial}{\partial z}n}{4\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}-\frac{3\,l m^2 n\frac{\partial}{\partial z}M\frac{\partial}{\partial z}n}{8\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}\\
&-\frac{l^2n^2\frac{\partial}{\partial z}M\frac{\partial}{\partial z}n}{8\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}-\frac{\sqrt{m^2+ln}\,m^2n\frac{\partial}{\partial t}M\frac{\partial}{\partial t}m}{4\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}-\frac{\sqrt{m^2+ln}\,l n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial t}m}{4\left(e^M m^4 n+2\,e^M l m^2 n^2+e^M l^2 n^3\right)}-\dots
\end{aligned}
$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"